



SQL - DDL Assignment

Task 1 : Create Table

Create a table named **Students** with the following columns:

Column	Data Type	Constraints/Notes
student_id	INT	Primary Key, Not Null
name	VARCHAR(50)	Not Null
gender	CHAR(1)	Check: value in {'M','F'}
age	INT	Check: age > 0
course	VARCHAR(50)	Default: 'BCA'
email	VARCHAR(100)	Nullable

Task 2 : Add Rows

Add 10 rows to the table created with the following info:

student_id	name	gender	age	course	email
1	Aarav Sharma	M	20	BCA	aarav@example.com
2	Ananya Iyer	F	21	BBA	ananya@example.com
3	Rohan Mehta	M	19	BSc	rohan@example.com
4	Isha Kapoor	F	22	BCom	isha@example.com
5	Vikram Rao	M	20	BCA	vikram@example.com
6	Neha Verma	F	23	BSc	neha@example.com
7	Arjun Nair	M	18	BA	arjun@example.com
8	Sara Khan	F	21	BBA	sara@example.com
9	Kabir Singh	M	24	BCom	kabir@example.com
10	Priya Das	F	22	BCA	priya@example.com

Questions (Q1–Q10) : DDL Commands

For each question, examine the target output structure and write the DDL needed to reach it from the Base Table.

Q1. Output Table

Column	Data Type	Constraints/Notes
student_id	INT	Primary Key, Not Null
name	VARCHAR(50)	Not Null
gender	CHAR(1)	Check: value in {'M', 'F'}
age	INT	Check: age > 0
course	VARCHAR(80)	Default: 'BCA' (length increased)
email	VARCHAR(100)	Nullable

Q2. Output Table

Column	Data Type	Constraints/Notes
student_id	INT	Primary Key, Not Null
name	VARCHAR(50)	Not Null
gender	CHAR(1)	Check: value in {'M', 'F'}
age	INT	Check: age > 0
course	VARCHAR(50)	Default: 'BCA'
email	VARCHAR(100)	Not Null, Unique

Q3. Output Table

Column	Data Type	Constraints/Notes
student_id	INT	Primary Key, Not Null
name	VARCHAR(50)	Not Null
full_name	VARCHAR(60)	(new column added)
gender	CHAR(1)	Check: value in {'M', 'F'}
age	INT	Check: age > 0
course	VARCHAR(50)	Default: 'BCA'
email	VARCHAR(100)	Nullable

Q4. Output Table

Column	Data Type	Constraints/Notes
student_id	INT	Primary Key, Not Null
name	VARCHAR(50)	Not Null
gender	CHAR(1)	Check: value in {'M', 'F'}
course	VARCHAR(50)	Default: 'BCA'
email	VARCHAR(100)	Nullable

Q5. Output Table

Column	Data Type	Constraints/Notes
student_id	INT	Primary Key, Not Null
name	VARCHAR(50)	Not Null
gender	CHAR(1)	Check: value in {'M', 'F'}
age	INT	Check: age > 0
course	VARCHAR(50)	Default: 'BCA'
email	VARCHAR(100)	Nullable
phone	VARCHAR(15)	(newly added column)

Q6. Output Table

Column	Data Type	Constraints/Notes
student_id	INT	Primary Key, Not Null
name	VARCHAR(50)	Not Null
gender	CHAR(1)	Default: 'F', Check: value in {'M','F'}
age	INT	Check: age > 0
course	VARCHAR(50)	Default: 'BCA'
email	VARCHAR(100)	Nullable

Q7. Output Table

Column	Data Type	Constraints/Notes
student_id	INT	Primary Key, Not Null
name	VARCHAR(50)	Not Null
gender	CHAR(1)	Check: value in {'M', 'F'}
age	INT	Check: age > 0
course	VARCHAR(50)	Default: 'BCA'
email	VARCHAR(100)	Nullable
created_at	DATETIME	Default: CURRENT_TIMESTAMP

Q8. Output Table

Column	Data Type	Constraints/Notes
student_id	BIGINT	Primary Key, Not Null
name	VARCHAR(50)	Not Null
gender	CHAR(1)	Check: value in {'M', 'F'}
age	INT	Check: age > 0
course	VARCHAR(50)	Default: 'BCA'
email	VARCHAR(100)	Nullable

Q9. Output Table

Column	Data Type	Constraints/Notes
student_id	INT	Primary Key, Not Null
name	VARCHAR(50)	Not Null
gender	CHAR(1)	Check: value in {'M', 'F'}
age	INT	Check: age > 0
course	VARCHAR(50)	Default: 'BCA'
email	VARCHAR(100)	Nullable

Q10. Output Table

Column	Data Type	Constraints/Notes
student_id	INT	Primary Key, Not Null
name	VARCHAR(50)	Not Null
gender	CHAR(1)	Check: value in {'M', 'F'}
age	INT	Check: age > 0
course	VARCHAR(50)	Default: 'BCA'
email	VARCHAR(100)	Nullable